

Differential Calculus

*Home work sheet #1

Limit

Find the limit of the following functions:

1. $\lim_{x \rightarrow 0} \frac{x}{\sqrt{x+1}-1}$

2. $\lim_{x \rightarrow 2} \frac{2x^2 - 5x + 2}{5x^2 - 7x - 6}$

3. $f(x) = \begin{cases} x^2 + 1 & , x > 0 \\ 1 & , x = 0 \\ 1 + x & , x < 0 \end{cases}$

4. $f(x) = \begin{cases} 3x - 1 & , x < 1 \\ 3 - x & , x > 1 \end{cases}$

Find $\lim_{x \rightarrow 0} f(x)$

Find $\lim_{x \rightarrow 1} f(x)$

5. $\lim_{x \rightarrow 0} \frac{x}{|x|}$

6. $\lim_{x \rightarrow \infty} \frac{3x + 5}{6x - 8}$

7. $f(x) = \begin{cases} 2 - x & , x < 1 \\ x^2 + 1 & , x > 1 \end{cases}$

8. $f(x) = \begin{cases} e^{\frac{-|x|}{2}} & , -1 < x < 0 \\ x^2 & , 0 < x < 2 \end{cases}$

Find $\lim_{x \rightarrow 1} f(x)$

Find $\lim_{x \rightarrow 0} f(x)$

9. $f(x) = \begin{cases} \frac{1}{x+2} & , x < -2 \\ x^2 - 5 & , -2 < x < 3 \\ \sqrt{x+13} & , x > 3 \end{cases}$

10. $f(x) = \begin{cases} x^2 & , x < 1 \\ 2.4 & , x = 1 \\ x^2 + 1 & , x > 1 \end{cases}$

Find $\lim_{x \rightarrow -2} f(x)$ and $\lim_{x \rightarrow 3} f(x)$

Does $\lim_{x \rightarrow 1} f(x)$ exist?

11. $\lim_{x \rightarrow \infty} (\sqrt{x^6 + 5x^3} - x^3)$

12. Prove that $\lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right)^x = e$

13. $\lim_{x \rightarrow \infty} \sqrt[3]{\frac{3x+5}{6x-8}}$

14. $\lim_{x \rightarrow -\infty} \frac{4x^2 - x}{2x^3 - 5}$

15. Let $f(x) = \begin{cases} 2x+1 & , x < 1 \\ 3-x & , x > 1 \end{cases}$, find $\lim_{x \rightarrow 1} f(x)$.

**These problems are for the students only as home work. Search the reference books for more examples.*